A Comparison of Hemodialysis Access Revision Rates by Access Type in ESRD Patients

Spencer Matthews - 16 March 2022

Contents

Abstract **1**

1 Introduction **1**

2 Methods **2**

2.1 Data 2

2.2 Statistical Modeling 4

3 Results **6**

3.1 Descriptive Statistics 6

3.2 Inferential Analyses 8

3.3 Predictive Analysis 9

3.4 Secondary Analyses 10

4 Discussion **10**

Appendix **11**

A Missing Data Analysis and Imputation 11

B Poisson Assumption Analysis 11

C Additional Interpretation of Parameters 13

D Additional Descriptive Statistics 13

E Model Search Results 14

Abstract

Data on over 1,500 dialysis patients in the United States were analyzed with the outcome of interest being the revision rate on dialysis accesses placed in the patient. Models were constructed to determine the effect of access type of revision rate. Both venous transposition arteriovenous fistula accesses and prosthetic graft accesses had higher revision rates than standard arteriovenous fistula accesses (68%, CI = (22%, 132%) and 93%, CI = (58%, 135%), respectively). No difference in revision rate was found between arteriovenous fistula accesses and prosthetic graft accesses. The difference in revision rate by access type is different across diabetic status, but not by age. Models were also constructed to accurately predict revision rates in a patient receiving their first access, and predictions presented.

1 Introduction

End Stage Renal Disease (ESRD) occurs when kidneys fail to function in a manner that can sustain life. When this occurs, patients must either be put on a treatment called hemodialysis or receive a transplant. In hemodialysis, a machine filters blood just as the kidney should do. This is a long and tedious treatment that must occur 3-4 times per week for many hours at a time, and consists of pulling blood out of the patient, running it through the machine, and then replacing it into the patient. Current estimates for the number of people in the United States with ESRD range between 450,000 and 800,000 (Trivedi et al., 2018; Kirchhoff, 2018).

With the frequency of hemodialysis treatments, using a regular needle in a vein is unfeasible and so permanent access points are placed on the patient. The first type of access is called a prosthetic graft, which is the cheapest to make and easiest to surgically attach to the patient. Due to the less expensive nature of these accesses and their difficulty to implement, they are more commonly used in rural areas or at less-developed medical centers. The second type of access is referred to autogenous arteriovenous fistula (AVF) of which there are two classes, standard attachment or a venous transposition. These accesses are more difficult to attach, and require a maturation period of 6-8 weeks before they can be used (Weiswasser et al., 2004).

The issue of concern here is that these accesses are not perfect, and many researchers are interested in the patency rate of the accesses. Patency refers to the access being viable, as these accesses often fail or require a revision. In fact, fewer than 50% of all access conduits remain patent for longer than three years (Trivedi et al., 2020). This is a huge healthcare cost, since anyone diagnosed with ESRD in the US has Medicare coverage by law, although it should also be noted that Medicare is not always the primary payer in all ESRD treatments (Kirchhoff, 2018). Aside from the monetary cost, the emotional and physical burden of having to create a new access is also very high for those patients and the people around them. Reasons for failures/revisions vary, but main ones include thrombosis (or blood clotting), infection, bleeding, and (in the case of AVFs) failure to mature (Hodges et al., 1997).

Previous research has had some success in showing that the AVF method is superior to the prosthetic graft in terms of patency (Astor et al., 2005; Trivedi et al., 2018), but studies have also shown that AVFs are associated with poor results in diabetics and older patients (Weiswasser et al., 2004). Other studies have analyzed the relationship between access failure and factors such as smoking, diabetes, and gender with mixed results (Hodges et al., 1997; Wetzig et al., 1985). One important note is the difference in vein size by gender, which plays a role in which access type is chosen (Marcus et al., 2007). However, it has been noted that many different reporting methods are used when determining what constitutes a "failure", which may impact the validity of results (Hodges et al., 1997).

In our study, we have data from a random sample of clinics across the United States. Patients were selected on their first visit, some initial demographic and medical information obtained, and followed for differential amounts of time with the number of revisions recorded (note that failures count as revisions). We will first perform an inferential analysis with the goals of (1) identifying and quantifying differences in revision rates between the access types and (2) determining if this association differs according to diabetic status and/or age. Then, we will perform a predictive analysis with the ultimate goal of predicting revision rates in common populations of people who are receiving their first access.

2 Methods

In our analysis of this data, we first considered the data and predictors of importance, and then came up with a modeling plan using common statistical techniques.

2.1 Data

The data we have are 1,583 individual patients who were randomly sampled from clinics across the United States. Since Medicare covers everyone who is on dialysis in the United States, we can assume that people who are receiving accesses for the first time are well-represented in this sample, and there are not any major sub-populations being under-represented. Thus, our population that we can generalize to is people in the United States getting an access placed for the first time, assuming that they are relatively within the same time frame as those in our sample (technologies hopefully improve over time so extrapolating too far into the future would not be desireable). The study is observational, and so caution needs to be taken when discussing the association between access type and revision rate in our inferential analysis. We now turn to the available data and its implication in our analysis.

- **Follow-up Time:** The amount of time (in days) that the patient was followed for. In modeling rates of revision, we will need to include this as an offset term so that it is adjusted for when interpreting covariates. There are 39 patients that had a follow-up time of zero, implying that the access failed the first day it was used. Since an offset of 0 is not possible in our chosen model, we added 1 to each follow-up time value. Thus, instead of follow-up time it is more a measure of the number of days used, starting at the first day. Due to the discrete nature of the follow-up time variable, it is difficult to precisely define this variable, but we believe adding one to each value is reasonable. Furthermore, without this change the Poisson assumption that as the interval of interest shrinks the probability of observing an event goes to zero does not hold. Additional analyses were performed to determine if this significantly changes our findings, and are presented in Section 3.4.
- **Number of Revisions:** The response in our analysis. It is a count of any kind of procedural revision or failure that occurred during the follow-up time.
- **Access Type:** The predictor of interest, and thus will be included in our models. Levels are: Prosthetic graft, standard AVF or venous transposition AVF. We will use standard AVF as the referent group in our analyses, as it is commonly thought to be the most effective.
- **Age:** A confounding variable that we will include in our model. As mentioned in the introduction, elderly patients do not do as well with a standard AVF accesses and so are more likely to have a different access type (the 6-8 week maturation period may also play a role in this). Furthermore, as you age the ability of your body to cope with changes is dimished, so it is natural that those who are older will have a higher revision rate.
- **Sex:** A precision variable. Females have smaller blood vessels and are therefore more likely to receive a prosthetic graft than males. However, the revision rate should not be inherently higher in women than in men. This variable will not be included in our inferential models, but will be considered in our predictive model.
- **Race:** A potential confounder variable that we will include in our model. Similar to sex, vein sizes differ across ethnic groups, and so race is associated with the type of access that is placed on the patient. Also, race is associated with the response through the channel of wealth. There is a large disparity of wealth between races which leads to better healthcare facilities and likely better placement/maintaining of accesses. This could contribute to an association between race and revision rate. Levels are Caucasian, African American, and Other. In our models we will use Caucasian as the referent class since it is the most common.
- **Smoking:** A potential confounding variable that we will include in our model. Smoking has been shown to impact revision rate, and those who smoke often have smaller veins which impacts the type of access they receive. Levels are Never smoked, past smoker, and current smoker. We will use never smoked as the referent group in our models. Note that this is self-reported and so may be inaccurate.

- **BMI:** A confounding variable that we will include in our model. BMI is associated with the type of access as those who are overweight tend to have smaller/harder to find veins. Furthermore, BMI is an indicator of health which will likely impact the chances for blood clotting that could require a revision. We note that in this dataset, there are 15 records with mis-typed BMI values. We assume this since BMI >125 implies that an average height person weighs over 405kg (890lbs). Although this is possible, the chance that we have 15 of them in our dataset is extremely small, so we felt comfortable altering the data points. We did this by dividing their given BMI by 10 and proceeding with the analysis.
- **Diabetes Status:** A confounding variable that we will include in our model. Again, diabetics have smaller veins which impacts the type of access that they receive, and there have been some studies showing that diabetics are more likely to have revisions to their accesses. Levels are Diabetic and non-Diabetic, and non-Diabetic will be used as the reference class.
- **Serum Calcium:** A nuisance variable that will not be included in our inferential models. This has not been shown to be significantly related to access revision, and it should not impact the type of access placed on the person.
- **Serum Phosphorous:** A nuisance variable that will not be included in our inferential models. This has not been shown to be significantly related to access revision, and it should not impact the type of access placed on the person.
- **Serum Albumin:** A potential confounder that we will include in our model. This is a marker of nutrition, and so will likely be related to access type through blood vessel size (i.e. if you are healthy you should have easier to access veins) and then of course people who are healthy will tend to have a lower rate or revision than an unhealthy population.

Overall, these factors listed do a good job of covering potential confounders with a couple of exceptions. This biggest one is some measure of economic status. This was brought up in the point on race, but that is not a great predictor. Presumably those who are more wealthy will have better access to improved medical care for initial placement and maintenance/monitoring. As mentioned in Section 1, AVFs are more expensive and require more skill to attach, so wealth will likely affect which access is given as well. Also, previous surgeries/operations along with medications could be confounding factors that are associated with which access is placed and the revision rate.

In addition to the problems with the data mentioned in the bulleted list, there was also a non-negligible amount of missing data. No data is missing in our response, predictor of interest, or offset variables. Proportion of missing data across the levels of the predictor of interest are shown in Table 1.

Table 1: Missing data patterns in predictors that we will include in our model.

	N (% of group)		
	S-AVF (N=407)	VT-AVF (N=113)	PG (N=1063)
Age	1 (0.25%)	0 (0%)	8 (0.75%)
Race	3 (0.74%)	1 (0.88%)	7 (0.66%)
Smoking	29 (7.13%)	9 (7.96%)	64 (6.02%)
BMI	5 (1.23%)	1 (0.88%)	44 (4.14%)
Diabetes	8 (1.97%)	3 (2.65%)	28 (2.63%)
Serum Albumin	27 (6.63%)	11 (9.73%)	87 (8.18%)

We see that on average, the proportion of missingness is similar across groups, with the possible exception of BMI that seems to have an unusually large proportion in the prosthetic graft group. There are also some patterns in data that is missing together. To solve these issues and not lose too much data, we will use mean Serum Albumin and BMI values to fill in those missing points, and then will perform linear interpolation to fill in missing values for Smoking Status and Diabetes. Patients with missing age and/or race will be discarded from our data set. After discarding those patients we are left with a sample size of $N = 1,564$. For further missing data analysis and details on the linear interpolation, see Appendix A.

2.2 Statistical Modeling

Due to the nature of the dataset and our response, we chose to perform Poisson regression to model the rate of access revision. In a Poisson regression model, we model the log of the mean response as a linear combination of predictors. However, in our dataset we have different amount of follow-up time for each individual, so that the mean response depends not only on the rate of revisions but also on the number of days the access was used.

$$\log(\mu) = \log(\lambda A) = \log(\lambda) + \log(A) \tag{1}$$

Thus, we model our mean, μ, as shown in Equation 1, where A is an offset term that corresponds to the number of days of use and λ is the rate of access revision. Since A is known, we can model $\log(\lambda)$ as a linear combination of predictors which allows us to interpret coefficients in the context of the rate of access revision.

When fitting the Poisson regression model, we assume that our data are distributed according to a Poisson distribution. It meets the criteria of a Poisson distribution, with the exception of the property that the expected value is equal to the variance. This property was checked by plotting the squared pearson residuals against the fitted values. These plots can be seen in Appendix B, and estimated dispersion parameters for our three models were all around 14. This indicates a huge departure from our assumed model, so we decided to use the robust (or sandwich) variance estimator for GLMs (Hampel et al., 2011). Although this requires that sample size be large, our sample of N = 1,564 after the removals described in Section 2.1 is definitely large enough for the asymptotic theory to apply, and we know that this variance estimator will return consistent estimates. That way, our inference will still be valid even though our assumed model does not hold. Plots of partial residuals were also examined to ensure that the functional form of our predictors was specified correctly.

The models were fit in `R` using the `glm()` function (R Core Team, 2021). The robust variance estimator was implemented in `R` by Dan Gillen in the function `glmCI()` and `robust.se.glm()` (Gillen, 2022). Code for these functions can be made available upon request.

2.2.1 Inferential Models

The first question we create a model to answer is whether or not there is a difference in access revision by access type. For this model, we include the confounders of age, race, smoking status, bmi, diabetes, and serum albumin in addition to our predictor of interest, access type, and our response, number of revisions, with the offset term being the number of days of use. This model is given in Equation 2, where $\mathbb{1}(\cdot)$ is the indicator function.

$$\begin{aligned}
\log(\mu) &= \log(\text{Revision Rate}) + \log(\text{Days of Use}) \\
&= \beta_0 + \beta_1 \mathbb{1}(\text{Access} = \text{VT-AVF}) + \beta_2 \mathbb{1}(\text{Access} = \text{PG}) + \beta_3(\text{Age}) + \\
&\quad \beta_4 \mathbb{1}(\text{Race} = \text{African American}) + \beta_5 \mathbb{1}(\text{Race} = \text{Other}) + \beta_6 \mathbb{1}(\text{Smoking Status} = \text{Past Smoker}) + \\
&\quad \beta_7 \mathbb{1}(\text{Smoking Status} = \text{Current Smoker}) + \beta_8(\text{BMI}) + \beta_9 \mathbb{1}(\text{Diabetes} = \text{Yes}) + \\
&\quad \beta_{10}(\text{Serum Albumin}) + \log(\text{Days of Use})
\end{aligned} \tag{2}$$

Confounders should be included in modeling so that we can be sure that the association we are seeing is the actual association between our predictor of interest and our response. Precision variables such as gender were not included because we want the marginal effect of access type on revision rate. These same principles apply when determining if there is a difference in the association between revision rate and access type across different levels of diabetes and/or age. We will give a brief interpretation of parameters of interest in this model and in subsequent models will also only give interpretations of parameters (or linear combinations of

parameters) of interest. For a complete interpretation of each parameter for the model in Equation 2, see Appendix C.

- We estimate that e^{β_1} is the relative difference in revision rate between the subpopulation with the venous transposition AVF and the subpopulation with the standard AVF, assuming that the two subpopulations are comparable in terms of age, race, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_2} is the relative difference in revision rate between the subpopulation with the prosthetic graft access and the subpopulation with the standard AVF access, assuming that the two subpopulations are comparable in terms of age, race, smoking status, BMI, diabetes, and serum Albumin.

Thus, to answer our question of interest for this model we care about e^{β_1} and/or e^{β_2} being significantly different from 1, and also about $e^{\beta_2-\beta_1}$ being significantly different from 1. We test the first differences through the Wald test, and test the third difference using a test for a linear contrasts, as implemented in `R` by Dan Gillen (Gillen, 2022). The model for determining if this difference exists across diabetic status is given in Equation 3.

$$\begin{aligned}
\log(\mu) &= \log(\text{Revision Rate}) + \log(\text{Days of Use}) \\
&= \beta_0 + \beta_1 \mathbb{1}(\text{Access} = \text{VT-AVF}) + \beta_2 \mathbb{1}(\text{Access} = \text{PG}) + \beta_3(\text{Age}) + \\
&\quad \beta_4 \mathbb{1}(\text{Race} = \text{African American}) + \beta_5 \mathbb{1}(\text{Race} = \text{Other}) + \beta_6 \mathbb{1}(\text{Smoking Status} = \text{Past Smoker}) + \\
&\quad \beta_7 \mathbb{1}(\text{Smoking Status} = \text{Current Smoker}) + \beta_8(\text{BMI}) + \beta_9 \mathbb{1}(\text{Diabetes} = \text{Yes}) + \\
&\quad \beta_{10}(\text{Serum Albumin}) + \beta_{11} \mathbb{1}(\text{Access} = \text{VT-AVF}) \mathbb{1}(\text{Diabetes} = \text{Yes}) + \\
&\quad \beta_{12} \mathbb{1}(\text{Access} = \text{PG}) \mathbb{1}(\text{Diabetes} = \text{Yes}) + \log(\text{Days of Use})
\end{aligned} \tag{3}$$

Here we would interpret $e^{\beta_{11}}$ as the relative difference in revision rate when comparing the subpopulation with diabetes and a venous transposition AVF access to the subpopulation without diabetes but also having a venous trasnposition AVF, assuming the subpopulations are comparable in age, race, smoking status, bmi, and serum albumin. Similarly, $e^{\beta_{12}}$ is the relative difference in revision rate when comparing the subpopulation with diabetes and a prosthetic graft access to the subpopulation without diabetes but also having a prosthetic graft access, assuming the subpopulations are comparable in age, race, smoking status, bmi, and serum albumin. The model to determine if there is a difference across age is given in Equation 4.

$$\begin{aligned}
\log(\mu) &= \log(\text{Revision Rate}) + \log(\text{Days of Use}) \\
&= \beta_0 + \beta_1 \mathbb{1}(\text{Access} = \text{VT-AVF}) + \beta_2 \mathbb{1}(\text{Access} = \text{PG}) + \beta_3(\text{Age}) + \\
&\quad \beta_4 \mathbb{1}(\text{Race} = \text{African American}) + \beta_5 \mathbb{1}(\text{Race} = \text{Other}) + \beta_6 \mathbb{1}(\text{Smoking Status} = \text{Past Smoker}) + \\
&\quad \beta_7 \mathbb{1}(\text{Smoking Status} = \text{Current Smoker}) + \beta_8(\text{BMI}) + \beta_9 \mathbb{1}(\text{Diabetes} = \text{Yes}) + \\
&\quad \beta_{10}(\text{Serum Albumin}) + \beta_{11} \mathbb{1}(\text{Access} = \text{VT-AVF})(\text{Age}) + \\
&\quad \beta_{12} \mathbb{1}(\text{Access} = \text{PG})(\text{Age}) + \log(\text{Days of Use})
\end{aligned} \tag{4}$$

In this model, we would interpret $e^{\beta_3+\beta_{11}}$ as the relative difference of revision rate in subpopulations that have the venous transposition AVF access but differ in age by one year, assuming the subpopulations are comparable in race, smoking status, BMI, diabetes, and serum albumin. Similarly, $e^{\beta_3+\beta_{12}}$ is the relative difference of revision rate in subpopulations that have the prosthetic graft access but differ in age by one year, assuming the subpopulations are comparable in race, smoking status, BMI, diabetes, and serum albumin. Thus, in both Equation 3 and 4, we care about determining if $e^{\beta_{11}}$ and/or $e^{\beta_{12}}$ are significantly different from 1. If they are, this will allow us to answer the question of interest in the affirmative.

2.2.2 Predictive Model

In addition to the inferential models, we also built a predictive model. We split our data into a test and train set, and then performed a search of the model space on the train set. The model search took a two-phased approach. First, we did a best subsets search on all main effects that were available to us. For each model, the naive MSE, AIC, BIC, and 10-fold cross validated MSE were recorded. This was performed in `R` using functions written by the author, but adapted from similar functions written by Dan Gillen, and code can be made available on request (Gillen, 2022). At this stage we assessed the results and selected a model of main effects, and then proceeded to test various interaction effects between the selected variables. Models were assessed in a similar fashion, and a final predictive model was chosen.

We used the final model to predict on the test set of data, to obtain an estimate of independent sample MSE. In addition, various common subpopulations were identified and predictions were made for those subpopulations. Included with predictions for these subpopulations we also computed approximate prediction intervals. This was done using the `predict()` function as implemented by the `trending` package in `R` (Schumacher and Jombart, 2021).

3 Results

In sum, the analysis yielded interesting and important results. We first discuss some descriptive statistics, then move into the results for our inferential and predictive analyses. Finally, a couple of follow-up analyses are discussed

3.1 Descriptive Statistics

In Table 2 selected data characterstics are presented both marginally and across the different access types. Note that these descriptive statistics are computed after the editing of the data as mentioned in Section 2.1.

We see that the majority of our sample is between 45 and 75, and that there are slightly more males than females in this data set. Most patients are Caucasian in Race but a large minority is African American, and a majority of our patients have diabetes even though only 10% of all Americans have diabetes. In Table 2 we can also see some of the confounders with relation to the predictor of interest. The type of access given to the patient appears to be associated with age, race, and diabetic status. There is also some evidence that BMI differs across access type, and although smoking is not uniformly associated, there does seem to be an association for current smokers.

Further description of this confounding is seen in Table 3. Here we see mild evidence of differences in revision rate across race, smoking status, and diabetic status. However, these differences are not huge, especially when considering the associated standard deviations so some of our initial assumptions on the confounding may be incorrect.

Figure 1 shows the relationship between potential confounders and the log of the daily revision rate, with patients with no revisions removed for exploratory purposes. We see that although each of these plots shows an overall trend, they all have significant variation that likely corresponds to large uncertainty in the trend. Thus, it seems that some of the confounders we decided on in our model may not be of concern.

We do note that in Table 3 we can see some potential interactions in our predictors. Especially in Diabetes, where we see that the venous trasnposition AVF rate difference is affected by whether or not the person has diabetes. Additional descriptive statistics dealing with potential interactions between variables can be seen in Appendix D. We also performed unadjusted poisson regression of access type on the access revision rate and found that when compared to standard AVF, the revision rates of venous transposition AVF and prosthetic graft were 76% and 102% higher, respectively. More details are given in Appendix D.

Table 2: Descriptive statistics of the dataset, marginal and split by access type (S-AVF = standard arteriovenous fistula, VT-AVF = venous transposition arteriovenous fistula, PG = prosthetic graft).

	N (Percent) or Mean (Std. Deviation)			
	S-AVF (N=403)	VT-AVF (N=112)	PG (N=1049)	All Types (N=1564)
Number of Revisions	0.7 (1.03)	0.96 (1.19)	1.38 (1.85)	1.17 (1.66)
Age				
30 and Under	32 (7.94%)	3 (2.68%)	35 (3.34%)	70 (4.48%)
31 - 45	62 (15.38%)	17 (15.18%)	96 (9.15%)	175 (11.19%)
46 - 60	101 (25.06%)	10 (8.93%)	236 (22.5%)	347 (22.19%)
61 - 75	149 (36.97%)	51 (45.54%)	475 (45.28%)	675 (43.16%)
Over 75	59 (14.64%)	31 (27.68%)	207 (19.73%)	297 (18.99%)
Sex				
Male	280 (69.48%)	68 (60.71%)	485 (46.23%)	833 (53.26%)
Female	123 (30.52%)	44 (39.29%)	564 (53.77%)	731 (46.74%)
Race				
Caucasian	259 (64.27%)	78 (69.64%)	601 (57.29%)	938 (59.97%)
African American	106 (26.3%)	25 (22.32%)	373 (35.56%)	504 (32.23%)
Other	38 (9.43%)	9 (8.04%)	75 (7.15%)	122 (7.8%)
Smoking Status				
Never Smoked	224 (55.58%)	67 (59.82%)	611 (58.25%)	902 (57.67%)
Former Smoker	110 (27.3%)	32 (28.57%)	318 (30.31%)	460 (29.41%)
Current Smoker	69 (17.12%)	13 (11.61%)	120 (11.44%)	202 (12.92%)
BMI	24.91 (5.76)	24.95 (6.1)	26.11 (6.61)	25.72 (6.39)
Has Diabetes	188 (46.65%)	54 (48.21%)	609 (58.06%)	851 (54.41%)
Serum Levels				
Calcium (mg/dL)	5.75 (1.89)	5.41 (1.85)	5.42 (1.86)	5.51 (1.87)
Phosphate (mg/dL)	8.68 (0.89)	8.46 (0.95)	8.62 (1.01)	8.62 (0.98)
Albumin (g/dL)	8.68 (0.89)	8.46 (0.95)	8.62 (1.01)	8.62 (0.98)

Table 3: Median (IQR) yearly revision rate by access type across levels of categorical confounding variables. (S-AVF = standard arteriovenous fistula, VT-AVF = venous transposition arteriovenous fistula, PG = prosthetic graft).

Characteristic	S-AVF (N=407)	VT-AVF (N=113)	PG (N=1063)
Race			
Caucasian	0 (1.28)	0.69 (3.68)	0.75 (2.93)
African American	0.7 (4.09)	1.31 (2.63)	1.01 (2.91)
Other	0 (1)	0 (2.13)	0 (1.72)
Smoking Status			
Never Smoked	0 (2.37)	0.77 (3.82)	0.81 (2.88)
Former Smoker	0 (1.92)	1.12 (2.08)	0.68 (2.25)
Current Smoker	0 (0.7)	0 (2.63)	1.19 (3.97)
Diabetes			
Yes	0.5 (2.37)	1.42 (3.75)	0.77 (2.53)
No	0 (1.34)	0 (2.83)	0.85 (3.28)

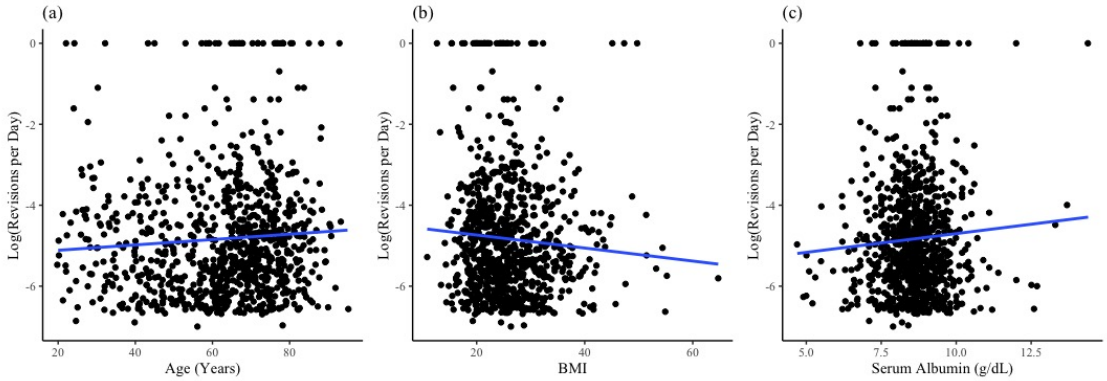

Figure 1: Plots showing relationship between potential confounders and the log of the daily rate of access revision, with those patients with no revisions removed. A linear fit is place on top of the points in order to see the overall trend. **(a)** Age shows a slightly positive relationship with revision rate **(b)** BMI shows a negative relationship with revision rate **(c)** Serum Albumin shows a negative relationship with revision rate.

3.2 Inferential Analyses

We fit the model shown in Equation 2 as described in Section 2.2.1. Parameter estimates along with 95% confidence intervals and corresponding p-values are given in Table 4.

Table 4: Estimated coefficients with associated confidence intervals and p-values for our primary inferential model

Coefficient	Exponentiated Estimate (95% CI)	P-Value
Intercept	0.002 (0.001, 0.004)	0
Access Type		
Standard AVF	Referent	
Venous Transposition AVF	1.682 (1.221, 2.317)	0.002
Prosthetic Graft	1.93 (1.582, 2.354)	0
Age	1.004 (0.998, 1.01)	0.158
Race		
Caucasian	Referent	
African American	1.116 (0.952, 1.308)	0.177
Other	0.69 (0.505, 0.942)	0.02
Smoking Status		
Never Smoked	Referent	
Past Smoker	0.982 (0.828, 1.165)	0.838
Current Smoker	1.162 (0.908, 1.486)	0.234
Diabetes		
No	Referent	
Yes	1.015 (0.863, 1.195)	0.854
BMI	0.999 (0.986, 1.012)	0.885
Serum Albumin	0.876 (0.765, 1.003)	0.056

Looking at this table, we estimate that the access revision rate is 68.2% (22.1%, 131.7%) higher in venous transposition AVF accesses than in standard AVF accesses, and 93% (58.2%, 135.4%) higher in prosthetic graft accesses than in standard AVF accesses. When we performed a test on the linear contrast of $\beta_1 - \beta_2$ to determine if there was a difference between venous transposition AVF and prosethetic graft accesses in terms of revision rate, we found that the difference was not significant. Our 95% confidence interval for the

relative difference between venous transposition AVF and prosethetic graft was between a 34% decrease and a 16% increase in revision rate.

After fitting the model shown in Equation 3, the interaction term between diabetes and having a venous transposition AVF was not significant. The interaction term between diabetes and having a prosthetic graft was however, Thus, we estimate that there is a 38.9% decrease in revision rate in the subpopulation of diabetics who have a prosthetic graft when compared to the subpopulation of non-diabetics who also have a prosthetic graft, assuming other factors are similar between the subpopulations.

Finally, when we fit the model shown in Equation 4, we saw that no interaction terms between the access type and age were significant. Thus, we can say that there is no difference in the association between access revision rate and access type across different ages.

3.3 Predictive Analysis

A search of the model space was conducted as described in Section 2.2.2. After the first stage, we used a combination of AIC and BIC to select our main effects, which resulted in the model including access type, age, race, and diabetic status. This model had an AIC of 5,331 and a BIC of 5,367. Further plots of the model search criteria against the number of parameters can be seen in Appendix E.

After analyzing predictive power of many models with interaction terms, we settled on the above listed main effects and additional interaction terms between access type and age, access type and race, and access type and diabetic status. This constituted our final model, which had an AIC of 5,316 and a BIC of 5,393. When tested on the independent test set, we found it to have a mean squared error of 2.15 when predicting revision counts.

Table 5: Predictions for access revision rate in various common subpopulation of people needing dialysis. Note that the revision rate here is in expected number of revisions per year. Included with the point estimate is an approximate 95% prediction interval

			Revisions per Year (95% PI)		
Age	Diabetic Status	Race	S-AVF	VT-AVF	PG
50	Non-Diabetic	Caucasian	0.404 (0.316, 0.517)	0.76 (0.478, 1.209)	1.272 (1.132, 1.429)
50	Non-Diabetic	African American	0.623 (0.479, 0.811)	0.657 (0.368, 1.172)	1.3 (1.151, 1.468)
50	Diabetic	Caucasian	0.584 (0.455, 0.751)	1.491 (1.005, 2.212)	1.236 (1.104, 1.385)
50	Diabetic	African American	0.901 (0.689, 1.178)	1.289 (0.763, 2.179)	1.263 (1.13, 1.412)
65	Non-Diabetic	Caucasian	0.519 (0.418, 0.644)	0.758 (0.532, 1.08)	1.327 (1.204, 1.463)
65	Non-Diabetic	African American	0.8 (0.609, 1.05)	0.655 (0.394, 1.089)	1.356 (1.207, 1.524)
65	Diabetic	Caucasian	0.75 (0.609, 0.923)	1.486 (1.11, 1.991)	1.29 (1.174, 1.416)
65	Diabetic	African American	1.156 (0.886, 1.508)	1.285 (0.805, 2.052)	1.318 (1.185, 1.465)
80	Non-Diabetic	Caucasian	0.665 (0.51, 0.867)	0.756 (0.512, 1.115)	1.384 (1.237, 1.55)
80	Non-Diabetic	African American	1.025 (0.73, 1.441)	0.653 (0.378, 1.128)	1.414 (1.229, 1.627)
80	Diabetic	Caucasian	0.961 (0.751, 1.231)	1.482 (1.03, 2.132)	1.345 (1.206, 1.5)
80	Diabetic	African American	1.482 (1.069, 2.056)	1.282 (0.755, 2.174)	1.375 (1.206, 1.567)

In Table 5 we see predictions for common subpopulations that are on dialysis (based on the dataset we have). Included with these predictions are approximate 95% prediction intervals. Note that these predictions are for the access revision rate per year, not the number of revisions, thus they can be interpreted as "we estimate that x (95% PI) is the revision rate (in revisions per year) for the subpopulation with the given characteristics." We see that as expected, predicted revision rate increases as age increases, and revision rate is higher for the prosthetic graft than for both types of AVF.

3.4 Secondary Analyses

In addition to the primary analyses, we performed some secondary analyses based on what we saw in the data. We performed a goodness-of-fit likelihood ratio test on the primary inferential model and found that the test was not significant, indicating that our model does a relatively good job of fitting the data. Note that in this test, we accounted for overdispersion by dividing our test statistic by our previous estimate of the dispersion parameter (about 14). Furthermore, we saw that in our primary inferential model, many of the predictors we chose did not have a large effect and thus should be removed to reduce model complexity. Specifically, we removed BMI, smoking status, race, and Serum Albumin from our model, and found that the likelihood ratio test between the smaller model and the larger model was not significant. This indicates that those predictors are likely not necessary in our model, although this is all post-hoc analyses and so nothing is certain. Again in this test we accounted for overdispersion. In summary, these two secondary analyses indicates that the models that we fit were unnecessarily complex. It also hints that the data fields we have available may not be the best at predictng revision rate. Future studies should seek additional demographic information that may be of more use in predicting revision rate.

Finally, we also fit the same primary inferential model but removed all records with a follow-up time of less than 14 days ($N = 66$). The motivating thought behind this being that perhaps we only care about people who have a viable access point to begin with, and we arbitrarily defined viable to be that it lasts for two weeks. In this analysis we found the relative difference in access rate revision for venous transposition AVF and prosthetic graft to be 74.5% and 100.3% higher, respectively, when compared to standard AVF accesses, assuming other factors are similar. Again, these numbers are similar to what we obtained in our primary inferential analysis and so we see that our data manipulation did little to impact the overall results.

4 Discussion

We found that there is a 68% increase in revision rate in venous transposition AVF accesses when compared to standard AVF, and a 93% increase in revision rate in prosthetic graft accesses when compared However, there is no significant difference in revision rate between venous transposition AVF accesses and prosthetic graft accesses. In general, this agrees with the literature that has a general consensus that standard AVF accesses are superior to prosthetic graft accesses. However, it is interesting to note that the venous transposition AVF access is not necessarily superior to the prosthetic graft. This is important because some patients cannot receive a standard AVF due to various physical limitations, and we cannot say that the costly venous transposition AVF would be better than a cheaper prosthetic graft. At this point interactions also come into play, as we determined that there is a significant interaction between access type and diabetic status. Those who have diabetes have a 39% lower revision rate with a prosthetic graft than those without diabetes. We are unsure as to why this occurs as it seems counter-intuitive. Further research will be needed in this area to describe more fully this relationship. We also determined that there is no interaction between age and access type, meaning that across different levels of age we should not give preference to one access type over another. However, this should be considered with the fact that due to physical limitations some elderly people cannot get AVF accesses.

We were also able to predict revision rate with a fairly high degree of accuracy, and create a table of predictions for common subpopulations in the hemodialysis community. These predictions could potentially be used to determine what kind of access would be optimal for a patient, as well as knowing what to expect in terms of follow-up care and future costs. Further research is needed to refine this model, and additional covariates may prove to be better predictors in the future. All the above results should be interpreted with the limitations set forth in Section 2.1. There are many missing values that do show some pattern which might be indicative of other factors at play. We have no socioeconomic data on these patients, which could be a large confounder that relates specifically to the quality of healthcare. In modeling, assumptions are made, and although we knew some assumptions were violated, steps were taken to ensure robustness against these violations. With these limitations and methods to address them in mind, we feel that it is reasonable to extend these results to the population of United States citizens who are receiving dialysis treatment for the first time.

Appendix

A Missing Data Analysis and Imputation

In addition to the univariate patterns of missingness noted in Section 2.1, we also analyzed two-dimensional patterns of missingness by looking at fields that were often missing together.

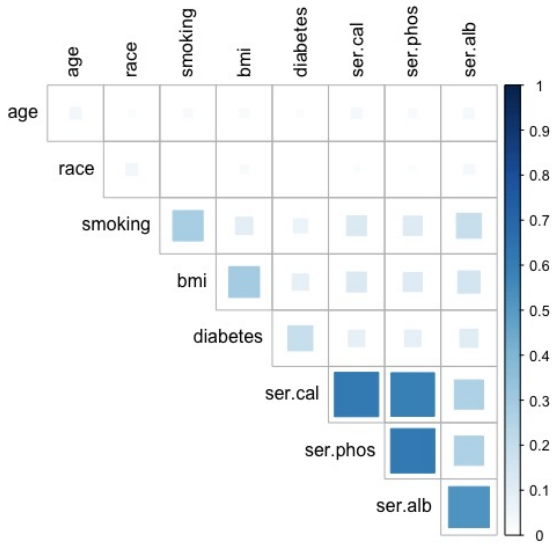

Figure 2: Visual representation of the probability of missing data occuring together between variables, given that at least two fields were missing for an individual.

These can be seen in Figure 2. The biggest pattern occurs between serum calcium and cerum phosphate, but we do not use either of those predictors in any of our models. We do note that Serum Albumin is often missing with other variables and is a predictor that we use in our inferential model.

As mentioned in Section 2.1 we used imputation to fill in missing values for the diabetes and smoking status variables. For both models, we used access type, age, race, bmi, and serum albumin as predictors to predict diabetes in the first model and smoking status in the second model. The model for diabetes was a logistic regression model fit using the `glm()` function in `R` and had an overall accuracy of 60.48%. For the smoking model, we used the `multinom()` function in the package `nnet` to fit a multinomial regression model (Venables and Ripley, 2002). This model had an overall accuracy of 54.17%. Code used to fit the data and replace missing values can be made available upon request. Although neither of these models have particularly stellar performance, it is better than guessing in both cases, and affects a relatively small portion of our data set.

In addition, we fit a model with all missing data removed to see how it would affect parameter estimates.

The resulting estimates are given in Table 6. We see that each estimate is within the associated confidence interval for the main analysis model as given in Table 4. Thus, we determine that doing imputation did not change the results in this paper.

B Poisson Assumption Analysis

As mentioned in Section 2.2, we estimated the dispersion parameter for our models and found it to be around 14.1 for each model. This shows a large departure from the assumption of a Poisson distribution, which takes the dispersion parameter to be 1. Thus, if we did not account for it then all of our inference would be invalid, with the actual standard errors being only 26% of what they should be. However, we caught this

Table 6: Estimated coefficients when missing data was simply removed from the data set with associated confidence intervals and p-values for our primary inferential model

Coefficient	Exponentiated Estimate (95% CI)	P-Value
Intercept	0.002 (0.001, 0.004)	0
Access Type		
Standard AVF	Referent	
Venous Transposition AVF	1.74 (1.23, 2.46)	0.002
Prosthetic Graft	2.057 (1.65, 2.565)	0
Age	1.004 (0.997, 1.01)	0.251
Race		
Caucasian	Referent	
African American	1.16 (0.973, 1.383)	0.098
Other	0.706 (0.497, 1.002)	0.052
Smoking Status		
Never Smoked	Referent	
Past Smoker	0.991 (0.824, 1.191)	0.92
Current Smoker	1.174 (0.899, 1.533)	0.239
Diabetes		
No	Referent	
Yes	1.018 (0.848, 1.222)	0.849
BMI	0.998 (0.983, 1.012)	0.758
Serum Albumin	0.907 (0.788, 1.044)	0.176

issue and were able to account for it as described in Section 2.2. Further evidence for this departure from the Poisson distribution can be seen in Figure 3.

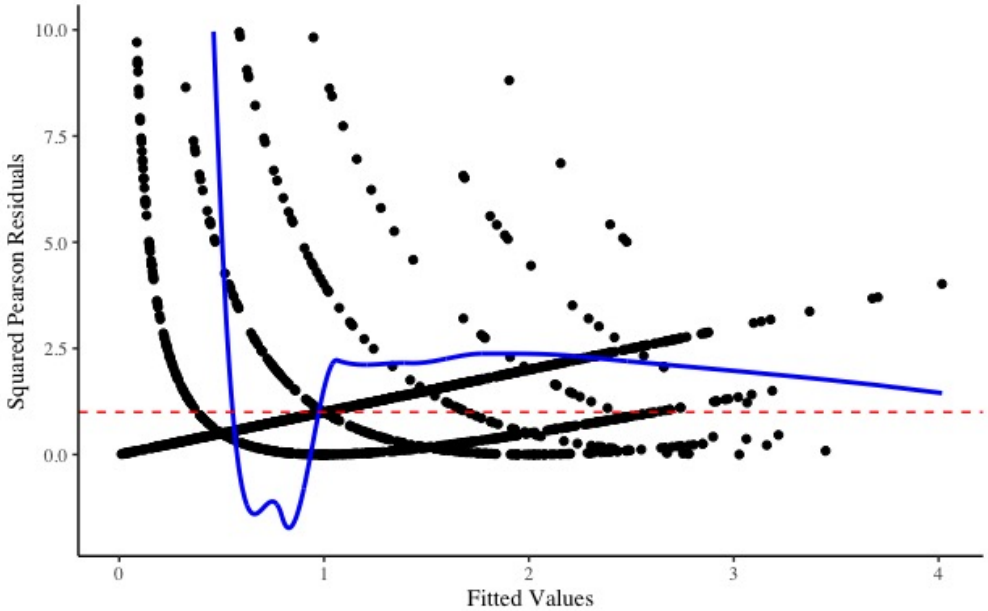

Figure 3: Plot showing squared pearson residuals vs. fitted values for the model described in Equation 2. The plot was similar for the other models, and the y-scale has been truncated so that the difference between the smoother (blue solid) and the theoretical line of 1 (red dashed) can be seen.

C Additional Interpretation of Parameters

The following are interpretations for each parameter in Equation 2:

- We estimate that e^{β_0} is the expected revision rate in the subpopulation of caucasian without diabetes who have never smoked, with a BMI of zero, a serum albumin of zero, and an age of zero, who have a standard AVF access. This subpopulation does not exist, so thi parameter is not practically interpretable for this model.
- We estimate that e^{β_1} is the relative difference in revision rate between the subpopulation with the venous transposition AVF and the subpopulation with the standard AVF, assuming that the two subpopulations are comparable in terms of age, race, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_2} is the relative difference in revision rate between the subpopulation with the prosthetic graft access and the subpopulation with the standard AVF access, assuming that the two subpopulations are comparable in terms of age, race, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_3} is the relative difference in revision rate between the subpopulation of a certain age and the subpopulation that is one year younger, assuming that the two subpopulations are comparable in terms of access type, race, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_4} is the relative difference in revision rate between the subpopulation of African Americans and the subpopulation of Caucasians, assuming that the two subpopulations are comparable in terms of access type, age, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_5} is the relative difference in revision rate between the subpopulation of Other races (not including African American) and the subpopulation of Caucasians, assuming that the two subpopulations are comparable in terms of access type, age, smoking status, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_6} is the relative difference in revision rate between the subpopulation of people who used to smoke and the subpopulation of people who have never smoked, assuming that the two subpopulations are comparable in terms of access type, age, race, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_7} is the relative difference in revision rate between the subpopulation of people who currently smoke and the subpopulation of people who have never smoked, assuming that the two subpopulations are comparable in terms of access type, age, race, BMI, diabetes, and serum Albumin.
- We estimate that e^{β_8} is the relative difference in revision rate between the subpopulation of people with a certain BMI and the subpopulation of people who have a BMI one unit less, assuming that the two subpopulations are comparable in terms of access type, age, race, smoking status, diabetes, and serum Albumin.
- We estimate that e^{β_9} is the relative difference in revision rate between the subpopulation of people with diabetes and the subpopulation of people without diabetes, assuming that the two subpopulations are comparable in terms of access type, age, race, smoking status, BMI, and serum Albumin.
- We estimate that $e^{\beta_{10}}$ is the relative difference in revision rate between the subpopulation of people with a certain serum Albumin and the subpopulation of people with on less g/dL of serum albumin, assuming that the two subpopulations are comparable in terms of access type, age, race, smoking status, BMI, and diabetes.

D Additional Descriptive Statistics

Here we discuss some additional descriptive statistics, especially in relation to interaction effects. In Figure 4 we see that there does appear to be a slight interaction between Age and access type, especially in the venous transposition AVF group. However, there are so few patients in this group it may not end up being significant (and in fact was not significant in the final model). Again in this figure, patients with no revisions were dropped for exploratory purposes.

In addition, we can also see an exploratory model fit for the unadjusted model in Table 7. These results are similar to the adjusted model, showing that the confounders we accounted for did not have a large impact on the effect of access type on revision rate.

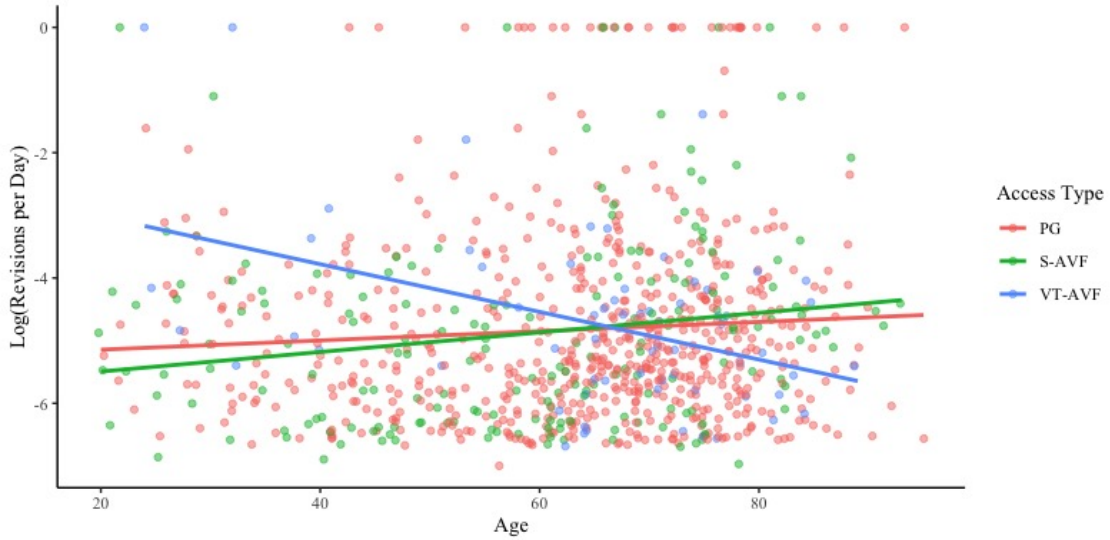

Figure 4: Plot showing the relationship between age and the log of the daily revision rate stratified by access type.

Table 7: Estimated coefficients with associated confidence intervals and p-values for the unadjusted model

Coefficient	Exponentiated Estimate (95% CI)	P-Value
Intercept	0.0017 (0.0014, 0.002)	0
Access Type		
Standard AVF	Referent	
Venous Transposition AVF	1.7626 (1.2967, 2.396)	3e-04
Prosthetic Graft	2.0257 (1.6848, 2.4357)	0

E Model Search Results

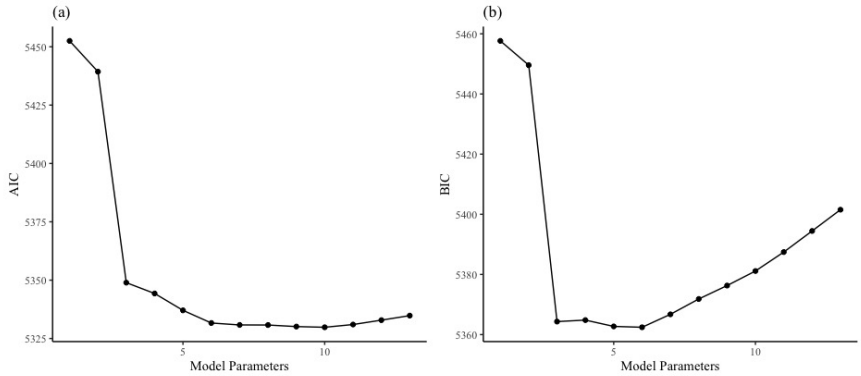

Figure 5: Models with best AIC (a) and BIC (b) for the main effects, with all number of parameters.

In Figure 5 we show the best model with each number of parameters for AIC and BIC. These were the main effect models, so no interactions are being fit here as that was done during the second stage of our predictive analysis. Note that AIC is minimized with 10 parameters in the model while BIC is minimized with only 6, as we would expect since the BIC penalty for additional parameters is larger than that of AIC.

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